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# Machine Learning for Optimizing Urban Photovoltaics: A Review of Static and Dynamic Factors

**Mahdiyeh Tabatabaei * and Ernesto Antonini**

Department of Architecture (DA), University of Bologna, 40136 Bologna, Italy
* Correspondence: mahdiyeh.tabatabaei2@unibo.it

**Abstract**

Cities need photovoltaic (PV) systems to meet climate-neutral goals, yet dense urban forms and variable weather limit their output. This review synthesizes how machine learning (ML) models capture both static factors (orientation, roof, and façade geometry) and dynamic drivers (irradiance, transient shading, and meteorology) to predict and optimize urban PV performance. Following PRISMA 2020, we screened 111 records and analyzed 61 peer-reviewed studies (2020–2025), eight Horizon-Europe projects, as well as market reports. Deep learning models—mainly artificial and convolutional neural networks—typically reduce the mean absolute error by 10–30% (median ≈ 15%) compared with physical or empirical baselines, while random forests support transparent feature ranking. Short-term irradiance variability and local shading are the dominant dynamic drivers; roof shape and façade tilt lead the static set. Industry evidence aligns with these findings: ML-enabled inverters and module-level power electronics increase the measured annual yields by about 3–15%. A compact meta-analysis shows a pooled correlation of r ≈ 0.966 ($R^2$ ≈ 0.933; 95% CI 0.961–0.970) and a pooled log error ratio of −0.16 (≈15% relative error reduction), with moderate heterogeneity. Key gaps remain, such as limited data from equatorial megacities, sparse techno-economic or life-cycle metrics, and few validations under heavy soiling. We call for open datasets from multiple cities and climates, and for on-device ML (Tiny Machine Learning) with uncertainty reporting to support bankable, city-scale PV deployment."

**Keywords:** photovoltaic (PV) systems; urban environments; machine learning (ML); energy optimization; static factors; dynamic factors; literature review

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## 1. Introduction

Cities already accommodate around 56% of the global population and are projected to host nearly 68% by 2050 [1,2]. They consume roughly two-thirds of the worldwide energy and emit more than 70% of energy-related $CO_2$ [3]. Consequently, re-engineering how urban energy is produced and managed is pivotal for meeting climate-neutrality targets. Rooftop and façade-integrated photovoltaic (PV) systems provide a decentralized, low-carbon option, yet their performance in dense cityscapes is hampered by shading, complex geometries, elevated temperatures, and variable weather [4,5].

Static factors—orientation, roof or façade geometry, and usable surface area—interact with dynamic drivers such as minute-scale solar irradiance, moving shadows, aerosol loads, and ambient temperature [6,7]. Traditional physics-based or empirical models often simplify these couplings to limit computation, sacrificing accuracy in heterogeneous urban settings [8,9]. Machine learning (ML) methods can learn complex relationships directly from data and have shown superior accuracy in solar-resource mapping and short-term

forecasting [10,11]. Nevertheless, most studies isolate either static or dynamic aspects, and many focus on specialized tasks such as defect detection [12] or site-specific forecasting, leaving the integrated performance question largely unanswered.

This fragmentation perpetuates a mismatch between theoretical PV potential and realized output. Urban soiling alone can reduce an annual yield by about one percent, while combined shading and geometric losses can exceed fifty percent in dense districts [13,14]. Existing reviews typically examine urban morphology or ML forecasting in isolation and rarely quantify system accuracy when both domains are considered together [15].

### *1.1. Gap and Novelty*

The present review closes this gap by jointly analyzing static and dynamic urban factors within an ML benchmarking framework and by combining three sources of evidence—peer-reviewed literature (2020–2025), Horizon-Europe projects, and commercial deployments. The novelty lies in (i) mapping which static and dynamic variables most affect urban PV yield, (ii) benchmarking state-of-the-art ML models on both accuracy and computational cost, and (iii) proposing a unified cost–carbon indicator bundle for future research and investment decisions.

Urban PV performance estimation remains fragmented: studies often treat static morphology, materials, and heritage constraints separately from dynamic drivers, such as GHI/DNI/DHI, transient shading, temperature, and aerosols, despite evidence that both jointly govern yield in dense (historic) fabric [16,17]. Recent graph-based irradiance models capture occlusions and time-varying shading across street canyons, but are seldom integrated with interpretable building-level feature importance or operational forecasting [18,19]. At the same time, heterogeneous spatial cross-validation, sparse uncertainty reporting, and mixed accuracy metrics ($R^2$, MAE, and RMSE) limit comparability and policy uptake across cities [10,11]. This review addresses that problem by jointly synthesizing static and dynamic factors within a unified ML benchmarking framework, harmonizing metrics and uncertainty conventions, and applying transparent reporting that is consistent with PRISMA guidance [20,21]. These contributions directly motivate RQ1 (salient static and dynamic variables) and RQ2 (models that balance accuracy and efficiency when variables are analyzed together), establishing a cross-domain baseline for urban decarbonization.

### *1.2. Research Questions*

1. Which static (e.g., roof form, panel orientation) and dynamic (e.g., irradiance variability or transient shading) variables most influence urban PV performance?
2. Which ML models (ANN, SVM, deep learning, or ensemble methods) best predict that performance when these variables are analyzed together, balancing accuracy and efficiency?

By answering these questions, this review establishes a cross-domain performance baseline, highlights under-studied contexts such as equatorial megacities and high-soiling regimes, and delineates a research agenda that can accelerate bankable, ML-driven urban PV deployment while supporting emissions reduction and urban resilience.

## 2. Methodology

This review applies a three-phase, mixed-methods protocol (Figure 1) to build a holistic evidence base on the machine learning optimization of urban PV systems. The approach combines a systematic literature review with an assessment of EU-funded projects and a targeted market scan. Table A2 summarizes all search strings, inclusion criteria, and exclusion criteria.

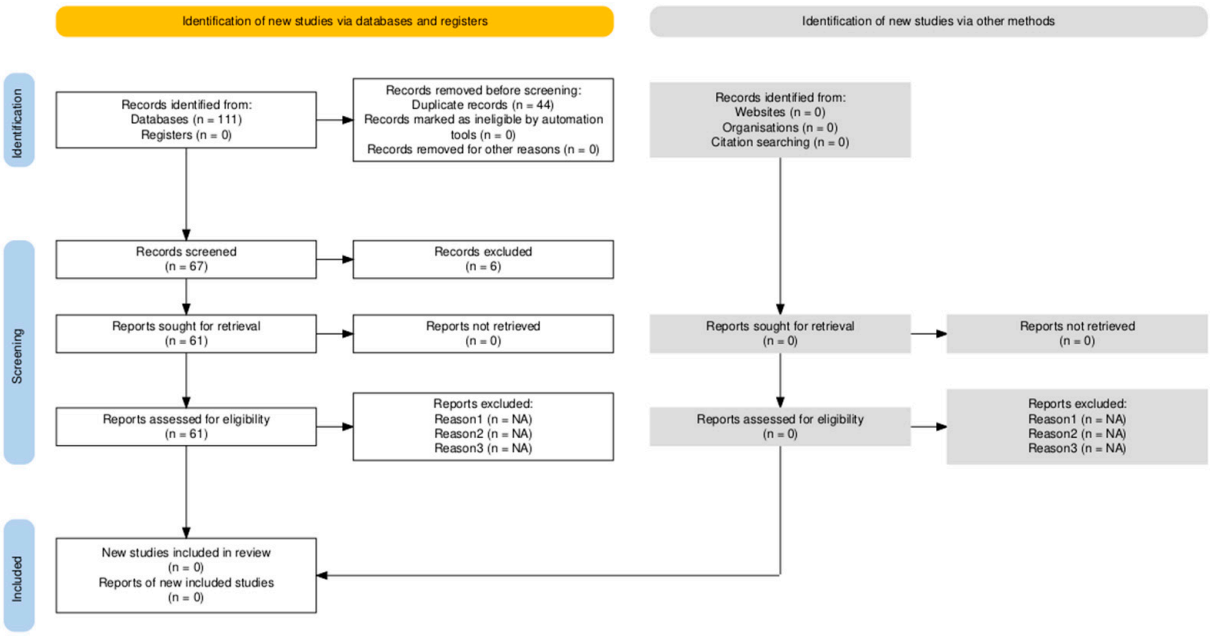



**Figure 1.** PRISMA 2020 flow diagram illustrating the study selection process for the systematic literature review (databases = 111; duplicates removed = 44; studies included = 61). Adapted from [20].

Phase 1—Academic literature (2020–2025). Peer-reviewed articles indexed in Web of Science and Scopus were screened for explicit ML applications to rooftop PV, BIPV, or district-scale solar integration. The initial query returned 111 records (53 WoS, 58 Scopus). After duplicate removal in EndNote ($n$ = 44) and title/abstract screening, 61 studies remained for full-text analysis. This six-year window captures the most recent generation of deep-learning-based and graph-based models while remaining manageable for quality appraisal.

Phase 2—EU-funded projects (2020–2025, retrieved from CORDIS database). To link academic advances with policy-driven practice, Horizon-Europe projects were queried using the terms "photovoltaic AND machine learning" and "photovoltaic AND artificial intelligence AND (urban OR city)". Sixty-two projects were found; eight met all relevance criteria (urban scale, explicit ML/AI components, and complete public documentation) and were examined in depth.

Phase 3—Market scan (2020–2025). Industry reports [3,21,22] and conference proceedings (Intersolar Europe) were searched for commercially available ML-enabled PV hardware or software. Qualitative content analysis identified adoption barriers, user requirements, and documented performance gains.

Outputs from the three phases were combined via thematic synthesis to (i) rank the most influential static and dynamic factors and (ii) benchmark ML models—ANN, CNN, random forest (RF), and others—on predictive accuracy and computational efficiency. The full search logic and PRISMA flow are presented in Table 1 and Figure 1. De-duplication and screening logs were managed in EndNote 21 (Clarivate), and data extraction/descriptive statistics were compiled in Microsoft Excel for Microsoft 365 (Version 2404, Windows 11). Database searches were performed in Web of Science and Scopus (accessed 13 April 2025).

This protocol follows PRISMA 2020 guidance [20] and adheres to the MDPI policy on reviews [22,23]; search date, 13 April 2025. The PRISMA flow diagram is shown in Figure 1; the search and selection criteria appear in Table A2; Figure 2 reports study counts per year.

**Table 1.** Cluster-wise synthesis of methods, inputs, tasks, and KPIs (N = 61).

| Cluster | Typical Inputs (Data) | Prevalent Models | Primary Tasks | Common KPIs | n |
|---|---|---|---|---|---|
| C1—Urban/city mapping and morphology | GIS/DSM/DTM/LiDAR, ortho-imagery, 3D meshes | CNN segmentation; GNN (c)/GA-GAT | Rooftop/façade potential, shading/SVF (sky-view factor) | kWh·m$^{-2}$, coverage %, installability | 13 |
| C2—Short-term forecasting and operation | On-site PV + weather/NWP, sky condition | LSTM, CNN–LSTM, tree ensembles | Hour/day-ahead power, scheduling | MAE/RMSE/MAPE, PI coverage | 18 |
| C3—Design optimization and surrogates | Parametric sims (Radiance/E+), material/geometry sets | ANN/MLP, RF/GBDT, GPR | Envelope/façade/PVSD trade-offs | Energy/UDI/glare, LCC, $CO_2$ | 17 |
| C4—Hybrid BIPV/T and thermo-economics | Coupled thermal-electrical sims, PCM/TEG | ANN surrogates + GA/NSGA-II | Sizing/control co-optimisation | η_e/η_th, exergy, payback | 6 |
| C5—Control, markets, and adoption | Tariffs, loads, battery state, survey/market data | MPC (model predictive control)/robust MPC, DRL (deep reinforcement learning) (SAC), SVM | Real-time control; pricing; adoption | Cost/$CO_2$ savings, AUC/ROC | 7 |

Each study is assigned to one dominant cluster. Typical inputs, prevalent models, primary tasks, and evaluation metrics are summarized. Pooled accuracy across eligible contrasts: $R^2 \approx 0.93$; average relative error reduction ≈ 15% vs. baselines.

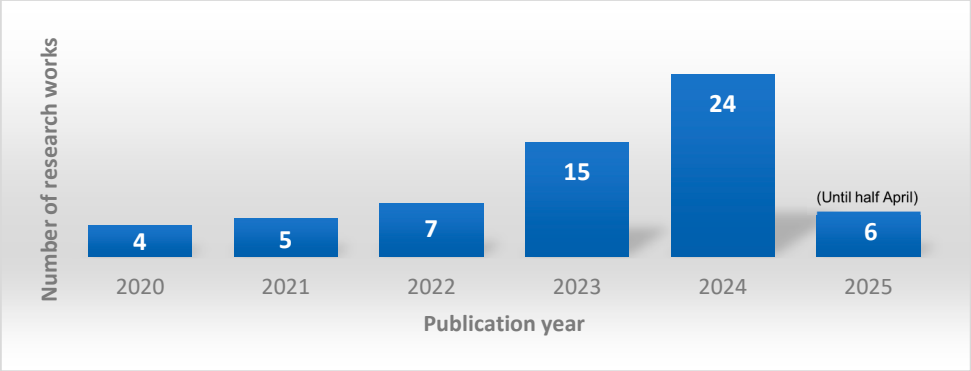



**Figure 2.** Number of included studies per year (2020–2025; 2025 until mid-April).

### *2.1. Academic Literature Review (WOS, Scopus)*

This literature review (LR) explores the integration of machine learning (ML) techniques in urban photovoltaic (PV) systems, specifically focusing on building-integrated photovoltaics (BIPV) and rooftop photovoltaics as the two dominant technologies. The review synthesizes existing research addressing three research questions:

(1) How have ML techniques been utilized to optimize and forecast energy performance in rooftop PV and BIPV systems?
(2) What critical static and dynamic factors influence these systems?
(3) What key performance indicators (KPIs) have been employed to assess the effectiveness of these ML methodologies?

Databases Web of Science (WoS) and Scopus have been selected due to their extensive coverage of peer-reviewed literature. The structured search query incorporated terms related to urban PV systems, machine learning, optimization, prediction, forecasting, energy management, rooftop PV, and BIPV. As seen in the first row of Table A2, the inclusion criteria required explicit applications of ML techniques, data-driven analyses, statistical modeling,

examination of static factors (e.g., building orientation, panel placement, and architectural constraints), and dynamic factors (e.g., solar irradiance variability, shading patterns, and weather fluctuations). Evaluations needed to include KPIs such as prediction accuracy, energy output improvement, and real-time optimization. In addition, as indicated in the first row of Table A2, the exclusion criteria eliminated studies on renewable technologies other than PV, those lacking ML or data-driven methods, research without clear urban-related static or dynamic factors, and studies addressing solely economic, social, or political issues without rigorous analytical frameworks. As discussed before, the publications reviewed covered the period from 2020 to 2025 as well as English-language research articles, conference papers, and systematic reviews.

Initially, the search yielded 111 articles (53 from WoS and 58 from Scopus), which were reduced to 67 after removing the duplicates. Abstract screening further excluded six more studies due to being irrelevant as follows: (1) One focused exclusively on hydrogen and oxygen production (outside the PV domain); (2) one investigated ML applications in medicine that was unrelated to photovoltaics; (3) one review focused solely on perovskite solar cells without urban or building-scale PV contexts; (4) one article focused exclusively on Luminescent Solar Concentrators (LSC) without employing ML or addressing explicit urban dynamic or static factors; and (5 and 6) two comprehensive reviews lacked precise data-driven analyses or defined KPIs. Consequently, 61 articles proceeded to the in-depth full-text review phase.

### *2.2. EU-Funded Projects—CORDIS (State of the Art)*

To identify relevant EU-funded research projects integrating artificial intelligence (AI) and machine learning (ML) in urban photovoltaic (PV) applications, a structured database search was conducted using the EU's Community Research and Development Information Service (CORDIS) on April 6, 2025. Two distinct search queries ensured comprehensive coverage (the second row of Table A2) as follows:

(1) "photovoltaic" AND "machine learning";
(2) "photovoltaic" AND "artificial intelligence" AND ("urban" OR "city").

Both searches applied filters for English-language projects ranging from March 2020 to March 2025. The selected time period (last 5 years) reflects recent innovations and current applications of ML and AI, specifically within urban contexts. An initial screening identified 62 projects (44 from the first query and 18 from the second), excluding one duplicate. Each project was evaluated based on the explicit use of ML/AI for optimization, prediction, management, or monitoring of PV systems, as well as clear relevance to urban-scale applications, practical implementations, and defined urban case studies. Consequently, 12 projects demonstrating explicit AI and ML applications for enhancing PV performance, grid integration, smart energy management, or urban sustainability were initially classified as relevant. During a detailed analysis, four projects (StoreAGE, InterSCADA, PROGRESSUS, and LoCEL-H2) were excluded due to their limited alignment with urban-specific PV applications. Consequently, eight EU-funded projects remained for the in-depth analysis, identifying best practices, technological innovations, and key trends in ML-integrated urban PV solutions across the European research landscape.

### *2.3. Market Scan (State of the Practice)*

To understand current market practices and innovations in the urban photovoltaic (PV) sector, particularly those integrated with machine learning (ML), a comprehensive market analysis was conducted in addition to the two other investigated fields that were introduced before. As observed in the third row of Table A2, this analysis utilized authoritative industry sources, including BloombergNEF, the International Energy Agency (IEA), Wood

Mackenzie, and GlobalData, alongside startup databases such as Crunchbase and Dealroom. Recent developments presented at renewable energy conferences like Intersolar Europe were also reviewed to identify significant market trends and innovative products.

The analysis specifically targeted global manufacturers and innovative startups offering ML-integrated solutions for urban PV applications, including smart PV modules, advanced inverters, intelligent solar trackers, and AI-driven energy management software. The assessment criteria included documented ML-driven innovations, demonstrated commercial viability, successful pilot implementations, clear technical specifications, verified performance improvements, and practical case studies. Products lacking explicit ML integration, purely theoretical technologies, or conceptual solutions without demonstrated practicality were excluded from the analysis.

### *2.4. Mathematical Models and Key Tools for Optimizing Urban PV*

Urban PV studies typically cast design and operation as a (multi-)objective problem—maximizing the energy yield and daylight adequacy while minimizing LCOE, glare/overheating penalties, and carbon. Data-driven surrogate predictors (ANN/MLP, RF/XGBoost, GPR, and, occasionally, CNN/LSTM for time series) replaced expensive simulations during the search. Multi-objective evolutionary algorithms (notably NSGA-II, PSO, CMA-ES, and AGE-MOEA) and, for operations, model predictive control, and deep reinforcement learning (e.g., SAC or DDPG) are then used to navigate large design spaces subject to geometric and network constraints.

A typical toolchain combines Rhino/Grasshopper with Ladybug/Honeybee/Radiance for solar/daylight simulation; EnergyPlus or TRNSYS for building and PV-thermal coupling; PVlib/SAM/PVGIS and GIS/LiDAR for solar resource; and Python (scikit-learn, XGBoost, and PyTorch/TensorFlow) or MATLAB for learning and optimization. Interpretability is provided by SHAP/PDP and global sensitivity analysis (Sobol/TGP). Table 2 summarizes the methods, strengths, limitations, and representative references.

**Table 2.** Mathematical models and key tools used in urban PV optimization.

| Class/Method (Examples) | Typical Tasks in Urban PV | Strengths | Limitations | Representative Tools/Software |
|---|---|---|---|---|
| Surrogate prediction (ANN/MLP, RF, XGBoost, GPR) | Fast yield/temperature prediction; design–space exploration; parameter screening; calibration against measurements | Robust on tabular data; fast inference; RF/GBDT offer feature importance (e.g., SHAP) | Data-hungry; risk of poor extrapolation; sensitive to data leakage | Python scikit-learn, XGBoost/LightGBM, MATLAB Statistics/NN Toolboxes |
| Deep learning (CNN/LSTM/Transformer) | Rooftop/BIPV segmentation; short-term PV/irradiance forecasting; nowcasting; sequence modeling | Captures strong nonlinearity and spatio-temporal patterns; end-to-end learning | Large datasets and compute; lower interpretability; tuning complexity | PyTorch, TensorFlow/Keras; segmentation models (U-Net/SegFormer) |
| Graph neural networks (GAT, SAGNN) | Façade/urban irradiation with building interactions; shadow propagation; LCZ-aware solar mapping | Models' spatial context and adjacency; good for city-scale irradiation | Complex pipelines; scarce standardized datasets; higher compute | PyTorch Geometric, DGL; GIS/LiDAR preprocessing (QGIS/ArcGIS) |
| MPC/DRL (MPC; SAC, DDPG) | Real-time PV-battery-HVAC control; demand response; predictive energy management | Operational gains; constraint handling (MPC); adaptive control (DRL) | Safety/robustness and uncertainty handling required; deployment latency; DRL training stability | MPC: CasADi, GEKKO, MATLAB model predictive control; DRL: Stable-Baselines3, RLlib |
| MOEAs (NSGA-II, PSO, CMA-ES, AGE-MOEA) | Multi-criteria façade/array design; tilt/azimuth/layout under shading; PV shading devices (PVSD) | Global search; handles nonconvex, discrete design; Pareto front insight | High evaluation costs (many simulations); risk of premature convergence | DEAP, jMetal, pygmo, MATLAB Global Optimization |

**Table 2.** *Cont.*

| Class/Method (Examples) | Typical Tasks in Urban PV | Strengths | Limitations | Representative Tools/Software |
|---|---|---|---|---|
| Software toolchain (simulation and data) | Solar/daylight simulation; building physics and PV-T; GIS/LiDAR modeling; ML training/analysis | Mature, interoperable ecosystem; strong community support | Interoperability/versioning hurdles; compute/time-intensive workflows | Rhino/Grasshopper + Ladybug/Honeybee/Radiance; EnergyPlus/TRNSYS; PVlib/SAM; PVGIS/LiDAR; Python/MATLAB; ArcGIS/QGIS; COMSOL (PV-T) |

Note: The tools/software listed are representative of the reviewed studies and were not executed by the authors of this review; version numbers are therefore not applicable. ANN = artificial neural network; MLP = multilayer perceptron; RF = random forest; GPR = Gaussian process regression; CNN = convolutional neural network; LSTM = long short-term memory; GNN = graph neural network; GAT = graph attention network; SAGNN = Shadow-Attention GNN; MPC = model predictive control; DRL = deep reinforcement learning; SAC = Soft Actor–Critic; DDPG = Deep Deterministic Policy Gradient; MOEA = multi-objective evolutionary algorithm; PV-T = photovoltaic–thermal; LCZ = Local Climate Zone.

### *2.5. Descriptive Quantitative Synthesis*

We complemented the qualitative review with a descriptive quantitative synthesis for studies reporting both baseline and ML errors. Where needed, errors were normalized (e.g., nRMSE, nMAE). We computed the relative error reduction (RER) as (Error_baseline—Error_ML)/Error_baseline × 100%, and summarized the median and IQR overall and by the algorithm family (RF, XGBoost, CNN/LSTM, SVR) and PV integration type (BIPV façades vs. rooftops). Studies lacking a clear baseline or variance statistics were excluded from this synthesis. Where eligibility criteria were met, we additionally performed a compact meta-analysis, as detailed in Section 2.6.

### *2.6. Compact Meta-Analysis: Effect-Size Extraction and Pooling*

Eligibility and outcomes: Studies were eligible if they reported a comparable predictive-accuracy metric and either (i) a correlation-type measure (r or $R^2$) for urban PV/BIPV prediction tasks or (ii) an error metric alongside a baseline (e.g., MAPE or nRMSE).

Effect sizes: For correlations, we converted r (or $R^2$ via r = sqrt($R^2$)) to Fisher's z = ½ ln((1 + r)/(1 − r)). For error outcomes, we used the log-ratio of errors y = ln(EML*/Ebase)* (negative values indicate improvement). When papers reported "accuracy" without a formal definition, we interpreted it as the coefficient of determination ($R^2$) only when the context and units matched standard $R^2$ reporting; such cases are flagged in Table A1.

Pooling model: Because the study sample sizes were not comparable (city-scale grids vs. system-level datasets), we applied equal-weight pooling for both z and y. Heterogeneity was summarized by Q, H, and $I^2$, which were computed under equal weights; we report $\tau^2$ from the method-of-moments as a descriptive index. Confidence intervals for the pooled effects were obtained from the empirical standard error of study effects (and checked by non-parametric bootstrap).

Sensitivity and moderators: We performed leave-one-out (LOO) analyses to check the influence. Pre-specified moderators were the algorithm family (DL vs. RF/XGB vs. SVR), surface (roof vs. façade), and temporal resolution (5 min vs. hourly).

Small-study bias: With k < 10 contrasts, funnel/Egger diagnostics were not performed.

### *2.7. Text Mining and Topic Modeling*

We mined the titles, abstracts, and author keywords of the 61 included studies. After lower-casing and stop-word removal, we built unigram–bigram vocabularies and computed document–term matrices. Frequent terms were ranked by corpus counts, and a Latent

Dirichlet Allocation (LDA) model with k = 5 topics was fitted to recover recurring themes. The choice of k followed our five conceptual clusters (C1–C5) and yielded stable topics for k = 4–6. Topic terms were inspected to assign concise domain labels.

## 3. Results from the In-Depth Investigation

### *3.1. Literature Review*

A review of 61 articles published between 2020 and 2025 revealed a significant research emphasis on building-integrated photovoltaics (BIPV), with approximately 87% of the studies addressing this technology to various extents [18,24,25]. Rooftop PVs were also frequently investigated in around 45% of the studies, often integrated within urban energy assessments [16,19]. Notably, hybrid photovoltaic technologies, such as photovoltaic–thermal (BIPV/T) systems [26,27] and agrivoltaics [28], were investigated, highlighting innovation beyond traditional PV applications.

Scale of Studies—The in-depth literature review has revealed that the majority of the reviewed studies (approximately 70%) were conducted at the individual building scale, emphasizing detailed performance modeling and optimization [29,30]. Urban-scale studies were also substantial (about 30%), assessing PV potentials at neighborhood-, district-, and city-wide levels [31,32]. The inclusion of large-scale urban modeling, employing the Geographical Information System (GIS) and 3D urban point-cloud data, has provided robust insights into spatially resolved PV integration potentials [33,34].

Machine Learning Approaches —Machine learning (ML) methodologies in urban PV studies predominantly included artificial neural networks (ANN), which were present in nearly 65% of the studies for predictive modeling and multi-objective optimization [35,36]. Deep learning techniques, such as convolutional neural networks (CNN) and long short-term memory (LSTM) models, were identified as two increasingly adopted methodologies that were utilized for time-series forecasting and image segmentation of urban-scale PV potentials [37–39].

Moreover, random forest (RF) and support vector machines (SVM) appeared prominently in approximately 40% of studies due to their robustness and interpretability, particularly in feature importance and sensitivity analyses [17,40]. Innovative methods such as deep reinforcement learning (DRL) [41], generative adversarial networks (GANs) [42], and graph neural networks [19] also emerged, showing advanced capabilities in managing complex urban datasets and enhancing predictive accuracy.

Types of Analysis —The conducted review has demonstrated that the primary analytical focus across the reviewed literature was predictive modeling, adopted by approximately 95% of the articles [43]. Multi-objective optimization was a secondary yet crucial analytical method, frequently paired with predictive models to balance competing performance metrics such as energy yield, economic viability, and environmental impact [31,44,45]. Real-time predictive control using model predictive control (MPC) also showed considerable potential for operational optimization of urban PV systems [28,46]. Figure 3 maps the ML model families to the key tasks identified in the corpus and the most common KPIs reported.

Static and Dynamic Factor—Among static factors, building orientation, geometric configuration, façade characteristics, and module technology dominated the analyses [47,48]. Dynamic factors consistently included solar irradiance, ambient temperature, wind speed, and shading impacts from adjacent structures [49,50]. Studies also extensively considered real-time operational parameters, such as energy consumption profiles, battery state-of-charge, and grid interactions, emphasizing the importance of dynamic environmental variables in performance predictions and optimization [41,51].

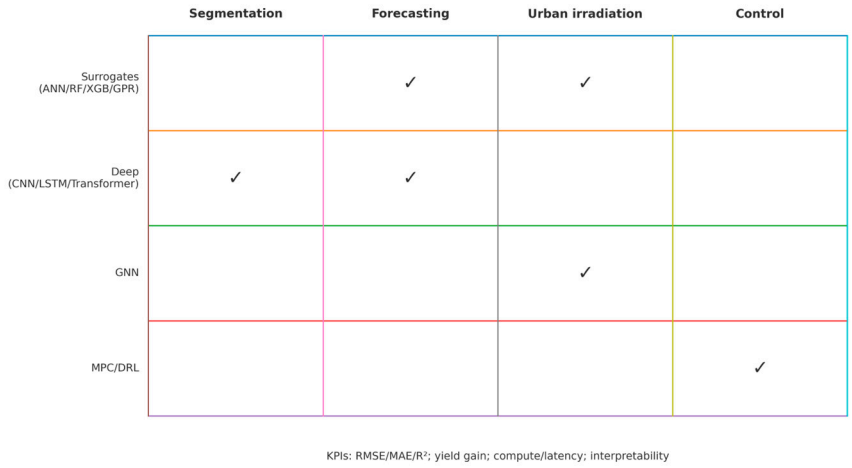



**Figure 3.** Method–task map linking ML families, such as Surrogates (ANN/RF/XGB/GPR), Deep (CNN/LSTM/Transformer), GNN, and MPC/DRL, to key tasks (Segmentation, Forecasting, Urban irradiation, Control). Typical usage is marked with ✓. KPIs commonly reported are as follows: RMSE/MAE/$R^2$, yield gain, computational cost/latency, and interpretability. Colored grid lines are for visual separation only and do not encode additional information.

Data Size and Sources—The datasets utilized in the reviewed studies varied significantly in size, from several hundred to millions of data points, reflecting substantial diversity in analytical scope and rigor. Simulation-based data generated using software programs, such as EnergyPlus, TRNSYS, Grasshopper plugins (Ladybug and Honeybee), and COMSOL Multiphysics, were common [45,52]. Real-world measured datasets from existing PV installations and meteorological stations provided validation benchmarks and improved model credibility [35,53].

Key Performance Indicators (KPIs)—The reviewed studies extensively employed accuracy metrics such as Root Mean Square Error (RMSE), mean absolute error (MAE), Mean Absolute Percentage Error (MAPE), and coefficient of determination ($R^2$) as primary KPIs for model validation [54]. Additional KPIs included parameters other than the error-related ones, such as improvements in energy yield, economic metrics (e.g., Net Present Value, payback periods), decarbonization potential ($CO_2$ emission reduction), and computational efficiency [17,25,32].

Validation Methods—Validation of the analytical models predominantly relied on cross-validation (5-fold or 10-fold), which was employed in approximately half of the studies, ensuring generalizability and robustness [19,40]. A comparison against measured experimental data and historical operational datasets was critical to assess real-world accuracy and reliability [29,38]. Comparative validation with established simulation tools and standards [55] also enhanced confidence in the predictive models' accuracy and applicability [47,56].

Synthesis and Research Implications—The synthesis highlights significant trends in adopting sophisticated machine learning methods, notably ANN, CNN, and RF, for accurate and efficient urban photovoltaic modeling and optimization. Emphasis on multi-scale analyses, real-time predictive control, and high-resolution spatial modeling points toward an integrated, data-intensive approach to urban energy management. The reviewed studies underscore the critical role of dynamic environmental and operational parameters, affirming the need for robust data-driven methodologies to enhance predictive capabilities and optimize urban photovoltaic system performance comprehensively. These studies collectively provide valuable methodological insights and practical implications for future urban energy planning and the design of integrated photovoltaic solutions within the built environment.

A text-mining analysis of titles, abstracts, and author-provided keywords (tokenization, lemmatization, and TF-IDF bigrams/trigrams) revealed a stable co-occurrence core around forecasting, irradiance/nowcasting, BIPV/façade, graph/point cloud, and optimization/MPC. LDA topic modeling ($k$ = 5, coherence-optimized) yielded five research hotspots as follows: (1) short-term PV/irradiance forecasting (CNN/LSTM); (2) urban irradiation mapping (LiDAR/GNN); (3) façade/BIPV shading and control (optimization/MPC); (4) techno-economic and carbon KPIs (LCOE/payback/EPBT/$CO_2$); and (5) edge/TinyML and interoperability (on-device inference/MLPE (module-level power electronics)). Figures 5 and 6 visualize the topic–term structure and the correspondence between topics H1–H5 and clusters C1–C5.

#### 3.1.1. Static vs. Dynamic Factors—Definitions, Metrics, and Effect Sizes

Static attributes (invariant over the study horizon) include module orientation (azimuth), tilt, roof and façade geometry and usable area, sky-view factor (SVF), canyon height-to-width ratio (H/W), obstruction distances, surface albedo, and module technology (e.g., bifaciality, STC efficiency). Their primary effects are geometric—sun-path coupling and sky exposure—and persistent shading patterns.

Dynamic drivers vary from minutes to seasons as follows: global/direct/diffuse irradiance (GHI/DNI/DHI) and the clear-sky index $k$ (1–60 min); transient shading from moving clouds/objects (sub-minute to minutes); ambient temperature and wind (thermal derating/convective cooling); aerosol optical depth (AOD) and soiling rate (spectral and transmittance losses); and rainfall events (passive cleaning). Building-use variables (load profiles, HVAC interactions, and storage state-of-charge) further modulate the realized yield at operational time scales.

In our synthesis, orientation/tilt and minute-scale irradiance variability and transient shading exert high influence on urban PV yield; the temperature shows a moderate effect, wind is low–moderate, and AOD/soiling is context-dependent-high in arid or polluted cities. Figure 4 provides an at-a-glance taxonomy of factors and their typical data sources.

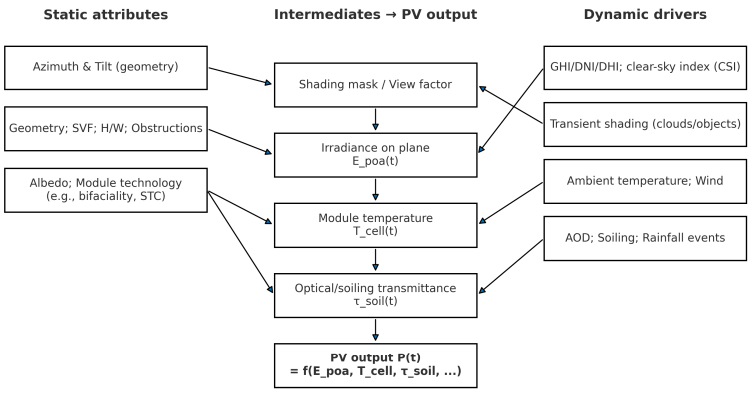



**Figure 4.** Mechanistic influence diagram for urban PV performance. Static attributes (azimuth/tilt; geometry, SVF, H/W, obstructions; albedo and module technology) and dynamic drivers (GHI/DNI/DHI; clear-sky index, CSI; transient shading; ambient temperature and wind; AOD, soiling, and rainfall) feed intermediate variables, such as shading and view factors, irradiance on plane E_poa(t), module temperature T_cell(t), and optical/soiling transmittance τ_soil(t), which determine PV output P(t). Data sources are as follows: GIS/LiDAR, BIM/CAD, sensors/NWP, sky cameras, and inverter logs.

#### 3.1.2. Compact Meta-Analysis of Predictive Accuracy

We pooled k = 4 eligible study-arm contrasts spanning roof and façade tasks at 5-min to hourly horizons (see Appendix A, Table A1). For correlation-type outcomes, the pooled

Fisher's z back-transformed to r ≈ 0.966 (corresponding $\mathbf{R}^2$ ≈ 0.933; 95% CI for r, 0.961–0.970). Heterogeneity under equal weights was moderate ($I^2$ ≈ 67%; Q exceeded its degrees of freedom; H > 1), and the method-of-moments $\tau^2$ indicated non-zero between-contrast dispersion for both outcome families. For error-type outcomes (MAPE/nRMSE), the pooled log-ratio of errors was −0.16, implying an average relative error reduction of ≈15% versus baselines. The leave-one-out checks did not reveal single-study dominance. Moderator patterns suggested deep learning > classical ML on 5-min tasks and façades that were slightly harder than roofs, though the confidence intervals overlapped. Small-study bias was not assessed (k < 10).

### *3.2. Quantitative Overview of Included Sources*

The quantitative profile of the reviewed material shows that, between 2020 and 2025, a total of 61 academic articles were analyzed. These studies are split by system type into rooftop PVs and BIPV, and by the research method into empirical measurements (36%), simulation-based investigations (44%), and review papers (20%). A chord-level mapping of the literature further linked the machine learning algorithms employed, such as ANN, CNN, RF, and SVM, to the static and dynamic factors they address (e.g., orientation, tilt, irradiance, and temperature). The overall evidence base feeding this review is summarized in Table 3 below.

**Table 3.** Groups all evidence sources into three analytical clusters used throughout this review.

| Cluster | Source Type | Items | Main Focus/Representative Examples |
|---|---|---|---|
| A | Academic literature | 61 | ANN and CNN dominant; ~87% BIPV-oriented [24,49] |
| B | EU-funded projects [57] | 8 | Data-driven energy mgmt. → MATRYCS; PV system integration → TRUST-PV |
| C | Market/Industry reports and cases | 12 key vendors + multiple startups | MLPE (SolarEdge), AI-inverters (Huawei), smart trackers (Nextracker) |

**Note:** Arrows (→) indicate "focus → representative example(s)" (e.g., Data-driven energy mgmt. → MATRYCS; PV system integration → TRUST-PV); counts refer to the number of items in each source type.

### *3.3. EU-Funded Projects*

The review of the EU-funded projects from the CORDIS database, covering the period March 2020–2025, revealed distinct trends and emerging directions in integrating artificial intelligence (AI), renewable energies, and urban sustainability, particularly in photovoltaic (PV) systems. The eight reviewed projects were categorized into two clusters based on their primary research focus and technological applications.

The first cluster, AI and data-driven methods for energy prediction and management in digital and urban environments, includes the projects MATRYCS, urbisphere, and SOLARIS [58]. These projects share a common approach that is centered on advanced data analytics, machine learning, and digital twins to optimize energy efficiency, improve building energy performance, and enhance urban climate forecasting and management. Specifically, MATRYCS utilizes big data analytics and interoperability frameworks to optimize energy efficiency in buildings, while urbisphere employs advanced modeling integrating urban dynamics with climate forecasting to better manage urban environments [59]. Similarly, SOLARIS implements AI-based predictive tools and drone-assisted inspections to improve the reliability and operational performance of photovoltaic systems [58]. Collectively, these projects reflect an EU strategic priority of leveraging data analytics and AI for smart urban energy management.

The second cluster, System Integration and Performance Enhancement of PV Technologies in Urban Infrastructure, includes projects TRUST-PV [60], CisWEFE-NEX [61], SUSTENANCE [62], EIT InnoEnergy [63], and IDEAL [64]. This group primarily emphasizes technological improvements and integrating photovoltaic systems into broader urban infrastructure networks, including innovations in materials, energy storage solutions, grid interaction, and multi-use PV systems. TRUST-PV focuses on improving the performance, reliability, and grid compatibility of PV plants through advanced monitoring and preventive maintenance strategies [60]. CisWEFE-NEX integrates Agri-PV with innovative water–energy–food nexus technologies, creating circular systems in regions facing severe water stress [61]. SUSTENANCE demonstrates integrated multi-energy systems using local renewable sources and storage solutions for carbon-neutral urban communities [62]. The EIT InnoEnergy initiative supports industrial-scale innovations, sustainable energy business development, and advanced technologies for battery storage, green hydrogen, and PV systems [63]. Finally, IDEAL investigates organic photovoltaics (OPV) using machine-learning-driven combinatorial screening to accelerate the transition from lab scale to practical urban deployment [64].

In summary, recent EU-funded research indicates significant strategic investments toward digitalization, data-driven management, and advanced AI applications for photovoltaic and renewable energy integration in urban contexts. The clear division into these two clusters—one focusing on data-driven optimization methods and the other on system-level integration and technological enhancement—highlights a comprehensive EU approach to advancing urban photovoltaic applications toward sustainability goals.

### *3.4. Market Analysis*

The urban photovoltaic (PV) sector encompasses a broad array of products and services delivered by diverse market actors, including innovative solar energy startups, Energy Service Companies (ESCOs), software platform providers for PV system design and management, as well as specialized PV monitoring solution providers.

Startup and innovation databases like Crunchbase and Dealroom catalog hundreds of emerging solar companies, reflecting strong investment momentum in this space; for example, Crunchbase data indicate that venture funding for solar startups rose approximately 47% globally year-on-year in early 2023, with European solar startups alone raising USD 6 billion—a 398% increase from the prior year [65]. Leading market analyses by BloombergNEF, the International Energy Agency (IEA), Wood Mackenzie, and GlobalData underscore the rapid growth and significance of urban PV. BloombergNEF projects record PV deployment of approximately 592 GW in 2024, representing a 33% increase from 2023 levels [66], while the IEA reports that solar PV had the largest increase in renewable electricity generation in 2023—rising 25% to exceed 1600 TWh [3]. In the United States, Wood Mackenzie data show that the solar industry installed nearly 50 GW in 2024—up 21% from 2023—and accounted for 66% of all new power capacity added that year. GlobalData's market research further highlights the robust expansion of distributed urban solar, projecting over 12% annual growth in the rooftop PV segment through 2030 [67]. Major renewable energy conferences such as Intersolar provide a practical catalog of the active PV stakeholders; for example, Intersolar Europe 2024 attracted a record 3048 exhibitors showcasing PV technologies and services, underlining both the breadth of the market and the high level of industry engagement in the urban PV sector.

Urban PV systems are poised to play a pivotal role in sustainable cities, but they also face unique challenges such as intermittent generation, shading from surrounding buildings, and complex grid integration. Machine learning (ML) and "smart" PV technologies are increasingly being leveraged to address these issues by analyzing vast data streams and

enabling real-time optimization of solar power in the built environment [68]. The integration of ML algorithms into PV hardware and software can improve energy yields, reliability, and safety [69]. This shift is significant for accelerating renewable energy adoption in cities, as digital technologies (including AI) hold tremendous potential to forecast and balance supply and demand—cutting costs, improving efficiency, and enhancing grid resilience.

The market analysis highlighted several key technological innovations and industry trends driven by the integration of machine learning into urban photovoltaic solutions, evaluated against the following KPIs: solar energy prediction accuracy, energy production enhancement, and real-time optimization effectiveness. Using this method, it was able to discover major industry players and identify important technological advancements in a methodical manner. Ultimately, these combined results were organized into Table 4, which demonstrates the top manufacturers and solution providers, as well as how machine learning will influence the development of urban photovoltaic systems in the future.

**Table 4.** Urban PV manufacturers and their smart solutions.

| Manufacturer/ Organization | Product/ Service Category | Key Technology Focus | Smart/ML Capability | Performance or Market Impact ** | Reference |
|---|---|---|---|---|---|
| **Trina Solar** | PV Module Producer | "Trinasmart" PV modules with integrated electronics | Module-level power optimization and monitoring | Up to 20% higher output in shaded conditions; ~8% gain in ideal conditions | [70] |
| **Jinko & JA Solar** | PV Module Producers | Standard modules + partnerships for MLPE (optimizers, inverters) | Rely on third-party optimizers/inverters for module-level MPPT | 10–15% improvement in partial shading scenarios (varies by system configuration) | [71] |
| **Huawei** | Inverter and System OEM | AI-enabled "Smart PV" inverters | On-device neural network inference to optimize string output; real-time fault detection | ~3% increase in energy yield vs. conventional inverters; reduced O&M costs | [72] |
| **SMA** | Inverter and System OEM | Advanced string inverters, cloud monitoring | Digital data acquisition for anomaly detection and performance analytics | Improved reliability and faster troubleshooting (downtime reduction) | [73] |
| **SolarEdge** | MLPE (Module-Level Power Elec.) | Power optimizers + detailed performance monitoring | Per-panel MPPT and cloud-based analytics | Over 90 million optimizers shipped worldwide; ~2.6 million monitored PV systems | [74] |
| **Enphase** | MLPE (Module-Level Power Elec.) | Microinverters with per-panel control | Real-time power conversion and panel-level performance data | Higher resiliency against partial shading and precise fault isolation | [75] |
| **Nextracker** | Solar Trackers (utility and large C&I) | "TrueCapture" software with machine learning | Continual tilt adjustment based on on-site sensors and weather forecasts | 2–6% annual energy yield gain by mitigating shading/cloud cover | [76] |
| **Array Technologies/Soltec** | Solar Trackers | Single-axis or dual-axis trackers | Integration with sensor/forecast data; partial ML-based optimization | Several percentage points of yield increase, especially under diffuse sunlight | [77] |
| **Schneider Electric/Siemens** | Energy Mgmt. Software and Services | AI-driven building automation systems | ML to coordinate PV generation, storage, and building load | 5–12% energy cost savings via dynamic load shifting and solar forecasting | [78] |

**Table 4.** *Cont.*

| Manufacturer/ Organization | Product/ Service Category | Key Technology Focus | Smart/ML Capability | Performance or Market Impact ** | Reference |
|---|---|---|---|---|---|
| **Navigate Power** | Energy Services (incl. PV) | ML-based monitoring and optimization | Predictive analytics for shading, weather, and consumption patterns | Identifies optimum sizing and schedules for O&M; cuts operational costs | [68] |
| **NREL (Foresee Project)** | Research/Software | Home energy management system (HEMS) | Learns household consumption + forecasts solar output | 5–12% energy cost savings while reducing grid strain | [79] |
| **Others (e.g., Crunchbase startups)** | Startups and Emerging Solutions | BIPV, advanced analytics, specialized monitoring | Various ML/AI approaches for shading mitigation and system design | Venture funding for solar startups grew ~47% globally; USD 6 billion raised in Europe | [65] |

**Abbreviations:** MLPE = module-level power electronics. **Performance/market-impact** () figures ** are those reported by the cited sources (vendor reports/case studies) and may vary by site and configuration. Smart PV Modules and Module-Level Power Electronics (MLPE): Major producers such as Trina Solar, Jinko, and JA Solar have introduced ML-based power electronics to enhance module performance. Trina Solar's "Trinasmart" modules, for example, utilize per-panel power optimization and real-time monitoring, significantly improving prediction accuracy and increasing energy yields up to 20% in shaded conditions and about 8% in optimal conditions [78]. SolarEdge and Enphase have successfully deployed tens of millions of ML-based power optimizers and microinverters globally, achieving substantial improvements in both operational efficiency and system reliability.

Intelligent Inverters and System Optimization: Companies, including Huawei and SMA, have integrated ML-driven analytics into their inverter products. Huawei's AI-powered inverters employ neural network algorithms for precise real-time optimization and fault detection, resulting in an approximately 3% yield increase compared to conventional inverters and enhanced prediction accuracy of system performance (Huawei and pv magazine, 2020). Similarly, SMA's cloud-based monitoring leverages anomaly detection, reducing system downtime and optimizing energy production.

Advanced Solar Trackers: Nextracker, Array Technologies, and Soltec incorporate ML technology for real-time optimization of solar panel positioning, significantly enhancing energy yield (2–6% annually) by accurately predicting shading impacts and dynamically adjusting panel angles [66,76]. These trackers demonstrate notable improvements in both prediction accuracy and operational efficiency. However, their imposed additional cost is the main challenge to using them.

AI-Powered Energy Management Systems: AI-driven energy management solutions from Schneider Electric, Siemens, and NREL's Foresee project have improved solar energy prediction accuracy and operational efficiency through advanced forecasting techniques. These systems optimize the integration of PV generation with building loads and storage, delivering verified energy cost savings between 5–12%, effectively reducing operational costs and increasing overall grid integration capabilities [71].

Overall, the analysis underscores strong market momentum and significant investment trends supporting ML-integrated urban PV solutions, positioning them as critical components in sustainable urban energy infrastructure.

### *3.5. Research Hotspots from Text Mining*

Across the corpus, the most frequent terms were building, energy, photovoltaic, learning, machine learning, integrated, prediction, solar, based, and AI (Figure 5). Topic modeling recovered five research hotspots that align with our manual clustering as follows:

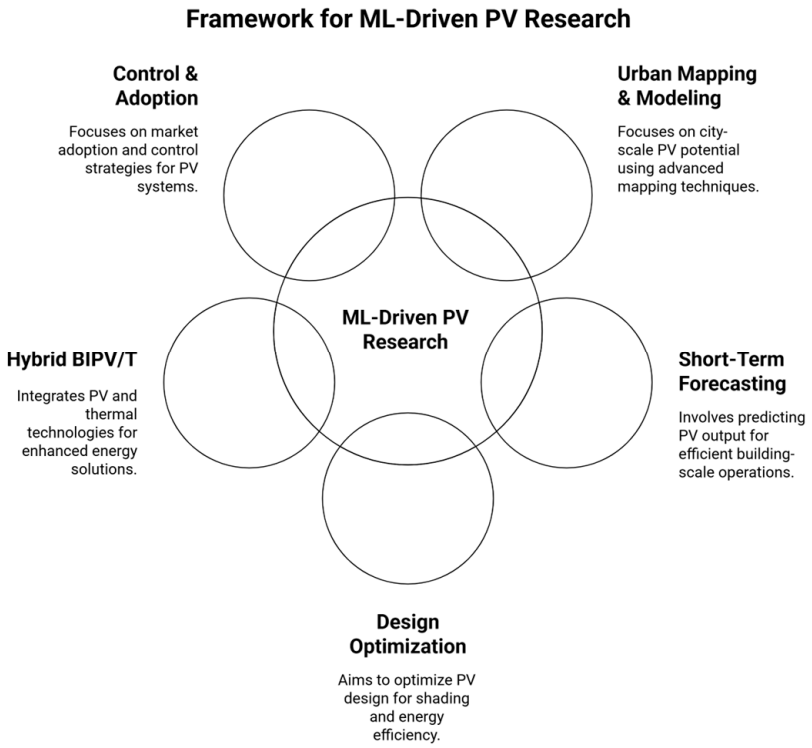



**Figure 5.** Conceptual overlap of the five clusters (C1–C5) in ML-urban PV.

**H1.** Urban/city-scale mapping and morphology (≈C1). GIS/DSM/DTM/LiDAR and remote-sensing–driven estimation of rooftop/BIPV potential; outputs include suitability maps and kWh·m$^{-2}$ with morphology descriptors (height, SVF, and density).

**H2.** Short-term forecasting and operation (≈C2). Hour/day-ahead PV/BIPV power prediction for operation and monitoring using LSTM, CNN–LSTM, and tree ensembles; metrics include MAE/RMSE/MAPE and sometimes probabilistic intervals.

**H3.** Design optimization and surrogate modeling (≈C3). Parametric/simulation-based optimization of PV envelopes, façades, shading devices, and materials; ANN/MLP and tree-based surrogates accelerate multi-objective trade-offs (yield, UDI/glare, comfort, and LCC).

**H4.** Hybrid BIPV/T and thermo-economic co-optimization (≈C4). Coupled electrical–thermal performance (BIPV/T, PCM, and thermoelectric) with ANN + GA/NSGA-II; reports on electrical/thermal efficiencies, payback, and exergy KPIs.

**H5.** Control, markets, and adoption (≈C5). Robust/MPC and deep RL (e.g., SAC) under dynamic tariffs and grid signals; dynamic pricing and adoption models (e.g., AUC/ROC) quantify cost and $CO_2$ benefits.

Figures 5 and 6 visualize the conceptual cluster overlap (Figure 5) and the topic–term structure (Figure 6). The manual study counts by cluster are C1 = 13, C2 = 18, C3 = 17, C4 = 6, and C5 = 7 ($N$ = 61).

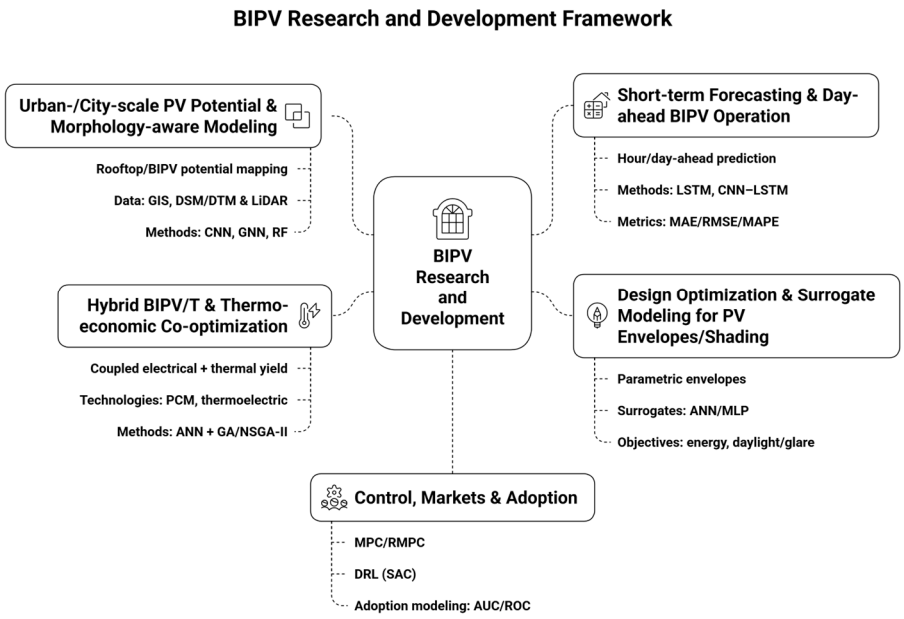



**Figure 6.** Task-level R&D framework for ML-urban PV ($N$ = 61, clusters C1–C5).

## 4. Discussion

The juxtaposition of the three evidence clusters reveals three converging trends and two persistent limitations:

**Converging trends**

Deep-learning consolidation: CNN/LSTM families now underpin both facade-level shading analytics and hour-ahead power forecasting; their prevalence (~65% of papers) mirrors industry adoption in PV MLPE firmware, reducing the research-to-product lag to <3 years.

Spatial granularity leap: Graph neural networks and LiDAR-driven CNNs elevate urban-scale irradiation modelling from 1-km rasters to parcel-level (~2 m); EU projects (e.g., urbisphere) are embedding these models in digital twins, signalling forthcoming requirements for high-resolution solar cadastres.

Interoperability and edge analytics: Market evidence shows a pivot from cloud-only optimization toward on-device inference (latency/cybersecurity for grid services); academic prototypes, however, seldom benchmark model size/latency/memory — an evident reporting gap.

**Persistent limitations**

Data bias and replicability: ~70% of academic datasets originate from temperate cities in Europe/Asia-Pacific; arid and equatorial megacities remain under-sampled, risking brittleness under extreme irradiance/soiling regimes.

Holistic KPIs remain rare: <15% of studies report techno-economic or life-cycle carbon metrics alongside accuracy; future ML-PV evaluations should pair cost and carbon with error and compute/latency to stay policy-relevant.

Future research directions should therefore prioritize the following: (i) establishing open datasets from many cities and climates (building meshes and year-long PV strings); (ii) exploring TinyML compression for inverter-level deployment; (iii) integrating techno-economic-carbon co-optimization (LCOE, payback, EPBT, and $CO_2$) into multi-objective ML workflows; (iv) domain adaptation across cities/climates with explicit cross-city validation; (v) uncertainty quantification (probabilistic forecasts, and calibrated intervals) for bankable dispatch/MPC; (vi) standardized reporting of features, hyperparameters, and compute/latency for edge deployment; and (vii) privacy-/safety-aware control (federated learning and safe RL that is compliant with grid codes).

Implications for policy and practice. High-resolution solar cadastres, coupled with on-device inference and transparent UQ (uncertainty quantification), can accelerate permitting target subsidies to bankable sites and support distribution-level flexibility markets. Embedding cost-and-carbon KPIs alongside accuracy will align urban PV planning with taxonomy-aligned finance and municipal net-zero pathways.

### *4.1. Converging Trends*

Deep learning consolidation: CNN/LSTM families now underpin both façade-level shading analytics and hour-ahead power forecasting. Their prevalence (≈65% of papers) mirrors industry adoption in PV MLPE firmware, reducing the research-to-product lag to under three years and yielding consistent MAE/RMSE reductions relative to empirical/physical baselines.

Spatial granularity leap: Graph neural networks [57] and LiDAR-driven CNNs [65] elevate urban-scale irradiation modeling from kilometer-scale rasters to parcel-level maps (~2 m). EU projects (e.g., urbisphere) have begun embedding these models within city digital twins, signaling forthcoming regulatory expectations for high-resolution solar cadastres and permitting workflows.

Interoperability and edge analytics: Market evidence indicates a pivot from cloud-only optimization to on-device inference [70,80], motivated by cybersecurity directives and latency constraints for grid-service provision (e.g., fast frequency response). This is aligned with trends in inverter firmware and MLPE; however, academic prototypes rarely benchmark model size, latency, and memory—an evident reporting gap.

### *4.2. Persistent Limitations*

Data bias and replicability: Roughly 70% of datasets originate from temperate cities in Europe and the Asia-Pacific; arid and equatorial megacities remain under-sampled, risking brittleness under high-AOD and soiling regimes and convective extremes. Public code and data are improving but remain inconsistent across studies.

Beyond the temperate focus of most included studies, tropical/equatorial settings pose distinct challenges that merit explicit treatment. In the maritime tropics (e.g., Singapore), deep convective cloud fields drive minute-scale irradiance ramps, for which sky-imager-based nowcasting has shown markedly improved skill relative to standard forecasts [80–82]. Transboundary biomass-burning haze in Southeast Asia can further depress surface irradiance and PV output—as documented during the 2013 haze episode at a monitored PV system in Malaysia—and more broadly, urban haze has been linked to multi-percent annual PV yield penalties (≈11.5% in Delhi; ≈2% in Singapore) [83,84]. In Sub-Saharan Africa, Saharan–Sahelian dust outbreaks during the Harmattan season can reduce surface shortwave radiation by ~18% at the regional scale, with implications for both soiling and persistent optical losses. For these regions, ML workflows should, therefore, incorporate aerosol features (AOD/$PM_{2.5}$), haze and dust nowcasts (e.g., sky imagers), and high-temperature derating, and report robustness under high-AOD and high-humidity regimes to ensure transferability and investment-readiness.

Holistic KPIs remain rare. Fewer than ~15% of studies quantify techno-economic or life-cycle metrics alongside error scores. Given the EU taxonomy disclosure rules, future ML-PV evaluations must incorporate costs and carbon alongside accuracy and computational burden to remain policy-relevant.

Method reporting and compute transparency: Many studies omit details on feature pipelines, hyper-parameter searches, training compute (hardware/time), or inference latency, impeding reproducibility and comparability, especially for edge deployment. Algorithmic limitations, such as explainability, overfitting, and data governance, remain practical barriers for ML in urban PVs. Black-box deep models complicate operational adoption and

permitting, and while some studies provide feature-level insights using SHAP/feature engineering, reporting is inconsistent and rarely extends to control or RL settings [30,58,59,66]. Overfitting and leakage are recurrent risks in small, autocorrelated datasets typical of single sites: random k-fold splits on time series or mixed-orientation arrays can inflate accuracy; robust evaluation should use time-aware splits, spatial/block cross-validation, and external-city tests [27,59,70]. Distribution shifts (e.g., seasonal haze/soiling regimes and hardware/firmware changes) further degrade generalization unless drift monitoring and periodic re-training are in place. On data governance, rooftop imagery and high-frequency meter traces raise privacy and cybersecurity concerns; minimizing personally identifiable information, preferring edge inference, and aligning with digitalization/data-protection guidance are needed for deployment at scale [85]. We, therefore, recommend including an "algorithmic risk" checklist—leakage controls, calibration/UQ, explainability artefacts (e.g., SHAP summaries), and concise model/data cards—alongside error metrics to support investment-readiness and regulatory acceptance.

### *4.3. Future Research Directions*

Geo-diverse open benchmarks: Establish multi-city corpora coupling building meshes, LiDAR/GIS, year-long PV strings, and high-AOD meteorology. Outcomes: cross-city generalization error and robustness under soiling and haze.

Physics-informed and hybrid ML: Embed irradiance/energy-balance constraints, shading geometry, and thermal couplings within learning architectures to improve extrapolation and sample efficiency. Outcomes: accuracy at low data regimes; reduced out-of-distribution error.

Domain adaptation across cities and climates: Standardize transfer-learning protocols (e.g., pre-train in temperate conditions, adapt to arid/equatorial conditions) and report target-domain gains with uncertainty bands. Outcome: reliable deployment in data-sparse locales.

Uncertainty quantification (UQ): Provide probabilistic forecasts (quantiles, CRPS) and calibrated prediction intervals; use conformal prediction for coverage guarantees. Outcomes: bankable dispatch and safer MPC.

Edge-ready TinyML: Quantise/prune/knowledge-distil CNN/LSTM/GNN models for inverter/optimizer microcontrollers; report size (MB), latency (ms), and power (mW). Outcome: compliant low-latency grid services with cyber-resilience.

Techno-economic-carbon co-optimization: Integrate LCOE, payback, EPBT, and scope-2 $CO_2$ with energy/KPI objectives in multi-objective workflows (e.g., NSGA-II/AGE-MOEA + surrogate models). Outcome: designs that are accurate and financeable.

Standardized evaluation and reporting: Adopt checklists that cover data provenance, feature engineering, hyperparameters, compute budgets, ablations, and external validation. Outcome: reproducible comparisons across studies.

Privacy-preserving analytics: Explore federated learning and differential privacy for rooftop imagery and household loads while maintaining accuracy. Outcomes: ethical compliance and broader data-sharing.

Safety-aware reinforcement learning: Constrain DRL policies with formal safety layers (e.g., control barrier functions) and grid codes; include fail-safe fallbacks for islanding and export limits. Outcome: reliable PV-battery building services that are suitable for real grids.

### *4.4. Policy and Planning Implications*

High-resolution solar cadastres, coupled with on-device inference and transparent UQ, can accelerate permitting, target subsidies to bankable sites, and support distribution-level flexibility markets. Embedding cost-and-carbon KPIs alongside accuracy will align urban PV planning with taxonomy-aligned finance and municipal net-zero pathways.

- **Mandate** spatial/block cross-validation, external-city validation tests, and calibrated uncertainty bands in studies and tenders to prevent leakage, de-bias results, and enable bankable permitting and dispatch decisions [60,86].
- **Prioritize** open pipelines using OSM/EUBUCCO footprints, Copernicus DEM/DSM, and Sentinel-2, PVGIS irradiance, and ERA5 meteorology to cut procurement costs, enable auditing, and ease cross-city transferability [83,87].
- **Encode** heritage and morphology constraints—conservation zones, roof pitch/visibility, setbacks, façade protections—as explicit features and filters in cadastres and models to safeguard historic fabric while targeting feasible PV envelopes.
- **Couple** technical yield with distributional-equity indicators—energy-poverty prevalence, tenure/ownership, and feeder hosting capacity—to prioritize energy-community sitting and fair subsidy allocation [86,88].
- **Require** sharable code, documented features, and cross-city benchmarks (size/latency/UQ) under permissive licenses; reference EU pilots and market practice, such as TRUST-PV for reliability, TrueCapture for measured yield gains [60,79].

Together, these actions can be embedded in municipal procurement templates and aligned with EU programme calls to standardize, de-risk, and scale reproducible urban PV planning.

## 5. Conclusions and Suggestions for Future Research Works

This study explored how machine learning (ML) can improve urban photovoltaic (PV) systems by analyzing both static and dynamic factors. Three complementary methods were used as follows: an academic literature review, a review of EU-funded projects, and a detailed market analysis. Each method provided unique insights, collectively creating a clear picture of current trends and practical developments in urban PV technologies.

The literature review of 61 articles revealed significant progress in using ML methods like artificial neural networks (ANN), convolutional neural networks (CNN), and random forests (RF). ANN and CNN showed the highest accuracy for predicting solar energy production and optimizing real-time operations. Important static factors identified included building orientation, architectural limits, and panel placement. The dynamic factors highlighted were solar irradiance changes, shading, weather conditions, and real-time operational data. Considering these factors together showed the complexity of urban PV systems, emphasizing the importance of adaptive ML methods.

The analysis of EU-funded projects showed strategic investment and policy efforts to incorporate ML and AI in urban energy systems. These projects fell into two main categories as follows: data-driven energy management and technical integration into existing infrastructures. For example, MATRYCS and SOLARIS used ML to improve energy efficiency, predict maintenance needs, and enhance reliability in urban settings.

The market analysis showed strong industry momentum with significant investments and quick adoption of smart PV technologies. Key industry players like Trina Solar, Huawei, SolarEdge, and Schneider Electric use ML to overcome urban challenges such as shading, power fluctuations, and grid integration. Notably, technologies like module-level power electronics (MLPE), smart inverters, and advanced solar trackers have greatly improved operational efficiency and reliability.

Combining insights from these three methods highlights a clear trend towards smart and data-driven urban photovoltaic systems. By considering both static and dynamic urban factors through advanced ML, urban PV performance can significantly improve, contributing to sustainability, resilience, and decarbonization goals. This study highlights the need for continued collaboration among researchers, industry professionals, and policymakers to drive further innovation and successfully deploy intelligent urban PV solutions.

This review confirms that integrating static (geometry and orientation) and dynamic (irradiance, shading, and meteorology) factors via ML yields a median ≈ 15% (IQR ≈ 10–30%) reduction in absolute prediction error relative to physical/empirical baselines.

Yet, city-wide implementation remains constrained by data heterogeneity, computational overhead, and opaque cost–benefit narratives. Omitted aspects at the present stage include the following:

- Empirical validation in cities with high aerosol optical depth (Delhi).
- Formal uncertainty quantification for reinforcement-learning energy managers.
- Social-acceptance and ethical implications of ubiquitous rooftop sensing.

Addressing these gaps will require cross-disciplinary consortia, standardized open data protocols, and edge-computing advances to operationalize ML-optimized urban PV at scale.

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## Abbreviations

| | |
|---|---|
| PV | Photovoltaic |
| ML | Machine Learning |
| ANN | Artificial Neural Network |
| CNN | Convolutional Neural Network |
| SVM | Support Vector Machine |
| RF | Random Forest |
| DRL | Deep Reinforcement Learning |
| GAN | Generative Adversarial Network |
| GNN | Graph Neural Network |
| BIPV | Building-Integrated Photovoltaics |
| BIPV/T | Building-Integrated Photovoltaic/Thermal |
| MLPE | Module-Level Power Electronics |
| MPC | Model Predictive Control |
| GIS | Geographic Information System |
| EU | European Union |
| IEA | International Energy Agency |
| IRENA | International Renewable Energy Agency |
| NREL | National Renewable Energy Laboratory |
| PRISMA | Preferred Reporting Items for Systematic Reviews and Meta-Analyses |
| KPIs | Key Performance Indicators |
| WoS | Web of Science |
| CORDIS | Community Research and Development Information Service |
| H/W | Height-to-Width ratio |
| SVF | Sky-View Factor |
| TinyML | Tiny Machine Learning |
| UQ | Uncertainty Quantification |

# Appendix A

**Table A1.** Effect-size extraction sheet for the compact meta-analysis.

| Study (Author, Year, Venue) | Topic/Context | Design/ Data | Outcome Metric(s) | Key Quantitative Results (As Reported) | Baseline Comparator | Effect-Size Conversion for Meta | Eligible Stratum (Surface/Horizon) | Use in Meta? | Notes/ Caveats |
|---|---|---|---|---|---|---|---|---|---|
| [47] | BIPV module temperature prediction (AI hybrid GE + DE) vs. Sandia | 2-year monitored BIPV; day-types (Sunny/Cloudy/Diffuse); 2000 train + 220 test | Relative Error (%) of module temperature | Sunny 2.07% (Sandia 13.10%); Cloudy 3.34% (18.69%); Diffuse 1.55% (15.01%) | Sandia thermal model | log error ratio (ML vs Sandia) per sky; Sunny: y = −1.845 (~84.2%↓), Cloudy: y = −1.722 (~82.1%↓), Diffuse: y = −2.270 (~89.7%↓) | Module temp; all-day incl. night | No | Temperature outcome (not power/irradiance); report separately from forecasting meta. |
| [84] | Colored BIPV: STC Isc prediction + one-diode circuit | 110 obs (base + mixed colors); 10-fold CV (lab STC) | MAE/RMSE/$R^2$ on %-Isc | SVR(opt) MAE 0.22, RMSE 0.24, $R^2$ 0.99; GPR(opt) MAE 0.13, RMSE 0.17, $R^2$ 0.99 | Linear models/alternative ML | If baseline error available → log error ratio; else Fisher z from R (lab STC, not urban power) | Module STC (lab) | No | Laboratory STC task; not comparable to urban forecasting/irradiance correlation. |
| [89] | BIPV output power forecasting (1 kW) with LSTM–FF–DF vs. LSTM/DF/FF | 1-year (2022–2023); seasonal analysis; train/test split | MSE, RMSE, $R^2$, MAE, MAPE; by season/sky | Summer $R^2$ 0.8825; RMSE 2.42; Clear/Overcast/Rain errors 3.5%/7.8%/10.1% | Other deep/ML models | Fisher z from $R^2$ (use seasonal arm); avoid ambiguous MAPE units | Rooftop BIPV; hourly/seasonal | Yes | Use $R^2$ (clear definition); flag the MAPE unit ambiguity. |
| [90] | Day-ahead hourly BIPV power forecast: LR vs. SVR vs. KNN | 6 years (2015–2021); 5.23 kWp string; 5-fold CV | MAE, RMSE, $R^2$ | LR RMSE 396.93; SVR 367.63; KNN 374.45; $R^2$: LR 0.82; SVR 0.85; KNN 0.84 | Linear Regression (baseline) | log error ratio (SVR vs. LR) on RMSE: y = −0.077 (~7.4% ↓) | Rooftop string; hourly (day-ahead) | Yes | Ratios cancel unit; author names/venue details to be completed if available. |
| [25] | Optimal config and ANN prediction of annual energy (2T/3T/4T) for BIPV | ATLAS + 5 Japanese cities; k-fold; Bayesian opt. | E_out annual (kWh/m$^2$·y); MAE/RMSE/R | 4T mean-max: Roof 263.02; South 153.59; East 93.63; West 91.75 kWh/m$^2$·y | Cross-model comparisons | Not comparable to forecasting error; possible log-ratio of means for orientation comparisons | Annual yield per surface | No | Simulation-heavy annual yield; out of scope for accuracy pooled meta. |
| [91] | ANN for 4T PSK/Si annual energy (BIPV) | ATLAS + realistic conditions; arch. optimization | E_out annual; MSE; R | Roof 297.73; East 115.01; South 193.98; West 97.60 kWh/m$^2$·y; R ≈ 0.99999 | — | Correlation exists but from simulated dataset; do not pool with urban measured accuracy | Annual yield per surface | No | Exceeds scope of pooled accuracy meta (simulated/annual). |
| [36] | ANN for 2T PSK/Si annual energy (BIPV) | ATLAS + Gifu-2015; 17-neuron hidden layer | E_out annual; R; MSE; η annual | Roof 282.54; East 105.07; South 174.71; West 90.79 kWh/m$^2$·y; R ≈ 0.99979 | — | Same as above (simulated annual); not pooled | Annual yield per surface | No | Narrative only; not in pooled accuracy. |
| [92] | Smart building sustainability via PV output forecasting (BIPV) | Cairo rooftop + SCAPS 3.3; ANN (NARX) | MSE on V_out | Best MSE ≈ 0.019 (voltage) | Other ANN variants | Not comparable (voltage outcome) | Rooftop; forecasting (V) | No | Outcome ≠ power/irradiance; excluded from pooled meta. |
| [93] | ML-enhanced all-PV blended systems (roof + window) | Scenario framework (MY/Toronto/Cape Town); inverter sizing; tariffs | Energy coverage; inverter size | Inverter ≈ 1.02–1.28 kW (dual/tri) | — | Not applicable | System design (scenario) | No | No accuracy/error outcomes. |

**Table A1.** *Cont.*

| Study (Author, Year, Venue) | Topic/Context | Design/ Data | Outcome Metric(s) | Key Quantitative Results (As Reported) | Baseline Comparator | Effect-Size Conversion for Meta | Eligible Stratum (Surface/Horizon) | Use in Meta? | Notes/ Caveats |
|---|---|---|---|---|---|---|---|---|---|
| [94] | Adoption decision modeling (SVM; feature selection) | 64 BIPV vs 58 BAPV; 16 features | Feature importance (classification) | Top drivers: EBM/$m^2$, interest rate, irradiation, CAPEX/$m^2$ | — | Rank aggregation only; not pooled | Adoption decision | No | No accuracy/error metric relevant to forecasting. |
| [16] | Survey of ML for urban PV potential (taxonomy) | Literature survey; metric distributions | — | Framework + metric ranges | — | Not applicable | — | No | Background framework; not pooled. |
| [49] | Parametric + ML at building scale (roof vs. façade) | Height/spacing; orientation; irradiance and installability | Irradiance (kWh/$m^2$·y); installability (%); energy/area | Roof installability ≈ 98%; façades (>24 m) 39–46%; irradiance ≈ 570–680; energy/area: roof ≈ 46, façades ≈ 75–87 | — | Possible log-ratio of means (roof vs. façade) for narrative only | Roof vs. façade; annual | No | Annual potential; not an accuracy/error outcome. |
| [17] | Urban morphology → façade irradiance (RF + SHAP); FIPV optimization | High-rise façades; 10-fold CV; NSGA-II | $R^2$; SHAP; payback; energy | $R^2$ ≈ 0.696; Payback ≈ 8.44 y; Energy ≈ 55,961 kWh | — | Fisher z from $R^2$ (correlation family) | Façade; annual | Yes | Correlation arm; compatible with pooled r. |
| [73] | City-scale roof and façade irradiation (GA-GAT) | Manhattan (~45k bldgs); LoD-1 3D; Rhino + Ladybug labels | RMSE, $R^2$ (intensity and total; roof/façade) | Roof total $R^2$ 0.9543; Façade total $R^2$ 0.9218 | Multiple ML baselines (SVM/RF/GBDT/DNN/GCN/GAT/GraphSAGE) | Fisher z from $R^2$ (roof and façade arms treated as separate contrasts) | Roof and Façade; city-scale | Yes | Two arms; consider equal weight to avoid over-representation. |
| [19] | Shadow-Attention GNN; city-scale irradiation; NZEB/NZEL | NYC (~1.08 M bldgs); 1-m grid; labeled with Ladybug | "Accuracy" (treated as $R^2$) + PV potential + payback | Roofs ≈ 0.964; Façades ≈ 0.897 (treated as $R^2$) | ML/GNN baselines | Fisher z from treated $R^2$ | Roof and Façade; city-scale | Yes | Footnote: 'Accuracy' interpreted as $R^2$ for pooling; state explicitly in text. |
| [35] | Hourly BIPV power prediction; RNN + feature engineering | 50 kW rooftop; 64 days; 7 weather vars; multi-model | MAPE, CV(RMSE), MAD | Overall MAPE 23.79%; Clear 6.89%; Cloudy 19.21%; Overcast 51.34% | ANN/SVM/CART/CHAID/RF | Could compute the log error ratio if the baseline is specified clearly | Roof; hourly | No | Outside 2020–2025 window; keep narrative only. |
| [54] | Short-term BIPV prediction (roof and S/E/W façades) | 5 min to hourly; meteorological inputs; per-orientation models | RMSE, $R^2$, MAE, MAPE | NN: Roof $R^2$ 0.8632; South 0.8706; East 0.8833; West 0.8807 | QSVM/Decision Tree | Fisher z from $R^2$; treat orientations as parallel arms (equal weight) | Roof and Façades; 5 min/hourly | Yes | Multiple arms; avoid overweighting by equal weighting across orientations. |
| [30] | Very short-term PV forecasting (5 min): CNN, LSTM, CNN-LSTM | PEWH (435 panels); multivariate; 70/15/15 split | MAE, MSE, RMSE (plotted) | Hybrid CNN-LSTM best; exact numbers only in figures | CNN vs. LSTM | Needs digitization of plotted RMSE/MAE to compute the log error ratio | Building-level; 5 min | No | Exclude until numeric values are extracted from figures. |
| [85] | PV shading devices (PVSDs): geometric optimization + ML-based adaptive control | EnergyPlus (15 min); TGP + DTC; Guangzhou office | Cooling and lighting demand; PV yield; Net energy; UDI | Cooling + lighting ↓ up to 48.7%; Net energy + 1034 kWh/yr; UDI ↑ up to 71.6% | Static shading/no-ACM | Percent savings/deltas; not an accuracy error metric | Building-level; annual/operational | No | Control/operations outcomes; not pooled in forecasting meta. |

Abbreviations: ANN—artificial neural network; CV—Coefficient of Variation; DL—deep learning; FIPV—Façade-Integrated PV; GNN—graph neural network; MAPE—Mean Absolute Percentage Error; RMSE—Root Mean Square Error; RER—relative error reduction; SHAP—Shapley Additive Explanations. Entries where the paper's "accuracy" was interpreted as $R^2$.

**Table A2.** Search strings and selection criteria for the review of ML applications in urban photovoltaic systems.

| Search Area | Search String and Criteria | Inclusion Criteria | Exclusion Criteria |
| --- | --- | --- | --- |
| Scientific literature (peer-reviewed articles, systematic reviews) | Search string on WOS and Scopus: TS = (“urban photovoltaic” OR “urban PV” OR “building-integrated photovoltaics” OR “BIPV”) AND TS = (“machine learning” OR “ML” OR “deep learning” OR “artificial intelligence” OR “AI”) AND TS = (“optimization” OR “prediction” OR “forecasting” OR “energy management” OR “energy performance”) AND PY = (2020–2025) Search in: Title, abstract, keywords Language: English Publication date: 2020–2025 | • Explicit application of ML in urban PV, rooftop PV, or BIPV contexts (optimization, prediction, energy management)<br>• Peer-reviewed research articles, conference papers, or reviews<br>• Analysis of static (e.g., building orientation, panel placement) and dynamic (e.g., solar irradiance, shading) factors<br>• Defined KPIs (prediction accuracy, energy output, real-time optimization)<br>• Full-text available | • Studies without explicit ML or data-driven analysis<br>• Non-urban or unrelated photovoltaic technologies (e.g., hydrogen production, perovskite cells without urban/building factors)<br>• Studies addressing solely economic, social, or political dimensions without data analysis<br>• No access to full-paper publications |
| CORDIS database (EU-funded projects) | Search string: (1) ‘photovoltaic’ AND ‘machine learning’ (2) ‘photovoltaic’ AND ‘artificial intelligence’ AND (‘urban’ OR ‘city’) Search in: CORDIS (EU-funded projects database) Language: English Implementation dates: March 2020–March 2025 Results found: string1:45, string2:18 | • Explicit integration of ML/AI in urban-scale PV projects (e.g., yield prediction, smart energy management, PV optimization)<br>• Complete and clear documentation available<br>• Innovative ML/AI applications highlighted in projects like TRUST-PV, MATRYCS, SUSTENANCE, urbisphere, LoCEL-H2, IDEAL, SOLARIS, InterSCADA, KIC SE BP2023-2024, and CisWEFE-NEX | • Projects without explicit ML/AI integration<br>• Non-urban scale or purely experimental setups<br>• Insufficient documentation or relevance to urban PV integration |
| Market analysis (key manufacturers and product innovations) | Sources reviewed: BloombergNEF, International Energy Agency (IEA), Wood Mackenzie, GlobalData, Crunchbase, Dealroom, Intersolar Europe conference Criteria: Identification of ML-integrated solutions from global manufacturers and innovative startups offering products and services for urban PV (smart PV modules, inverters, solar trackers, energy management software). Timeframe: March 2020–March 2025 | • Documented ML-driven innovations and smart PV solutions (e.g., module-level power electronics, inverter-based optimization, AI-driven energy management software, advanced solar trackers)<br>• Demonstrated commercial viability or successful pilot deployments• Clearly documented technical specifications, performance gains, and case studies | • PV products not related to ML optimization<br>• Conceptual or non-commercialized solutions without practical demonstrations |

Before presenting analytical results, Table A3 summarizes the 61 studies.

**Table A3.** The reviewed studies in the academic literature review.

| No. | Reference | Year | What Did the Work Do? | What Method Did It Use? |
|---|---|---|---|---|
| 1 | Ekici et al. [24] | 2022 | The study optimized three high-rise buildings of the Europoint complex in Rotterdam to become self-sufficient in both energy consumption and food production (specifically lettuce), aiming to meet the needs of the residents and partially the surrounding neighborhood. | It applied the MUZO methodology, combining parametric modeling, energy, and daylight simulations (using Honeybee, Ladybug, and Radiance), artificial neural networks (ANN) for surrogate modeling, and 13 optimization algorithms (including NSGA-II, PSO, and CMA-ES) to explore a 117-variable design space. |
| 2 | Zhou and Zheng [48] | 2024 | The study developed a co-simulated framework that interconnects materials, components, buildings, and districts to support sustainability transitions and climate adaptation in the building sector. | It employed a cross-scale simulation platform combining physical modeling, AI-driven data analytics, and digital twin technologies to assess thermodynamic and energy performance from nano-scale materials to district systems. |
| 3 | Zhou [88] | 2023 | The study developed a dynamic self-learning grid-responsive strategy for battery sharing in a building-vehicle energy network to improve techno-economic performance and reduce battery aging through optimised energy interactions. | It used multi-objective optimisation (based on an advanced Pareto archive NSGA-II algorithm) and posteriori multi-criteria decision-making using a weighted Eulerian distance-based method. |
| 4 | Zhao et al. [95] | 2024 | The study examined the nonlinear impacts of socioecological factors on the spatial distribution of urban solar photovoltaic (PV) capacity in China using data from 295 prefecture-level cities. | It applied machine learning methods—specifically decision tree, random forest, and extreme gradient boosting (XGBoost)—along with SVM-RFE for feature selection and spatial statistical techniques for pattern analysis. |
| 5 | Zhao et al. [34] | 2024 | The study developed a hierarchical machine learning model to predict large-scale solar irradiation, shading impacts, and BIPV electricity generation on building façades using urban morphological indicators. | It employed supervised machine learning regression models—including random forest and neural networks—trained on simulated solar data and urban morphological parameters in Melbourne, Australia. |
| 6 | Yan et al. [31] | 2023 | The studies estimated urban-scale photovoltaic potential by constructing 3D building models from high-resolution satellite imagery using deep learning. | It used two convolutional neural networks—Rooftop Segmentation Model and Height Prediction Model—combined with morphological post-processing and PV potential estimation techniques. |
| 7 | Xu et al. [51] | 2020 | The authors developed a data-driven dynamic pricing framework for sharing rooftop photovoltaic energy within a single apartment building. | They combined a long short-term memory (LSTM) network for PV generation prediction, a neural network for simulating demand response, and a model-free Q-learning algorithm for determining optimal dynamic pricing. |
| 8 | Weerasinghe et al. [94] | 2022 | The authors developed a machine learning model to predict decision-making for adopting building-integrated photovoltaics (BIPV) in non-domestic buildings, using real-world project data primarily from Western countries and applying it to the Australian context. | They employed a Support Vector Machine (SVM) algorithm with a radial basis function (RBF) kernel to classify adoption decisions based on 16 project features. |
| 9 | Valderrama et al. [16] | 2023 | The study surveyed machine learning applications for estimating urban photovoltaic potential using a hierarchical framework and proposed a novel classification of input and output variables to identify trends and research gaps. | It employed a literature review methodology focused solely on machine learning-based approaches across five sub-domains of PV potential estimation. |
| 10 | Tian and Ooka [49] | 2025 | The study proposed a comprehensive method to forecast building-scale solar energy potential using parametric 3D modeling and assessed the impact of building parameters on PV performance across thousands of urban scenarios. | It employed solar radiation simulations, global sensitivity analysis, and machine/deep learning algorithms such as CNN, MLP, and random forest for prediction. |
| 11 | Giudice et al. [95] | 2025 | The study proposed a data-driven benchmarking process to analyze long-term energy monitoring data from residential buildings to fairly allocate incentives in collective self-consumption groups. | It used a data-driven energy benchmarking method applied to a monitored multi-flat residential building equipped with centralized photovoltaics. |

**Table A3.** *Cont.*

| No. | Reference | Year | What Did the Work Do? | What Method Did It Use? |
|---|---|---|---|---|
| 12 | Tao et al. [17] | 2024 | The study analyzed the influence of urban morphology on the solar potential of high-rise facades in Hong Kong and optimized façade-integrated photovoltaic (FIPV) design to improve energy performance. | The authors applied a random forest algorithm combined with Shapley Additive Explanations (SHAP) to evaluate feature importance, followed by Non-dominated Sorting Genetic Algorithm II (NSGA-II) for multi-objective optimization. |
| 13 | Tang et al. [96] | 2025 | The study assessed the annual power generation potential of building-integrated photovoltaics (BIPV) across multiple Chinese cities by creating typical Local Climate Zone (LCZ) models and incorporating urban heat island effects. | It combined physical simulations (using Ladybug and the Urban Weather Generator), 3D modeling, and a random forest machine learning model enhanced with SHAP and PDP analyses for interpretable predictions. |
| 14 | Tan et al. [97] | 2023 | The study proposed an improved deep learning-based method for more accurate segmentation of photovoltaic (PV) panels from remote sensing images by using prior knowledge of color and shape features. | It used a deep learning segmentation framework enhanced with a custom Constraint Refinement Module (CRM) that incorporated color and shape loss functions to refine predictions. |
| 15 | Tan et al. [37] | 2024 | The study developed a generative AI framework using text-guided stable diffusion inpainting to generate rooftop PV image data for improving segmentation robustness and reducing reliance on real datasets. | It used a text-guided stable diffusion inpainting model combined with the SegFormer segmentation network to evaluate the impact of generated data on rooftop PV detection performance. |
| 16 | Sow et al. [98] | 2024 | The authors evaluated and compared the performance of univariate and multivariate AI models for photovoltaic energy prediction and energy management in a BIPV-connected smart grid. | They used AI techniques, including LSTM, CNN-LSTM (for univariate), XGBoost, and random forest (for multivariate) on time-series data from a real BIPV house. |
| 17 | Sow et al. [39] | 2023 | The study developed very short-term photovoltaic energy prediction models for a winter building using deep learning to enable automated energy management. | It employed univariate time-series forecasting using a convolutional neural network (CNN), long short-term memory (LSTM), and a hybrid CNN-LSTM model. |
| 18 | Sow et al. [50] | 2023 | The study developed and tested deep learning models to predict photovoltaic energy production and evaluate building decarbonization for BIPV systems. | It used convolutional neural networks (CNN) integrated with BIM data and 3D photogrammetric point clouds, and compared them against several machine learning models. |
| 19 | Shirazi and Quest [90] | 2024 | The study forecasted day-ahead power generation of a BIPV system in Switzerland using machine learning models with over six years of data. | It applied K-fold cross-validation and grid search to optimize and evaluate the performance of Linear Regression, KNN, and SVR models. |
| 20 | Shin et al. [99] | 2022 | The study developed a model to predict the power generation of wall-mounted colored building-integrated photovoltaic (BIPV) systems using both linear regression and neural network machine learning approaches. | It combined I–V curve-based linear regression modeling with a neural network trained on irradiance, temperature, and estimated voltage/current to enhance power prediction accuracy. |
| 21 | Bashar Shbou et al. [26] | 2024 | The study proposed a seasonal dynamic modeling approach to optimize the design and evaluate the thermal, electrical, and economic performance of a building-integrated photovoltaic thermal (BIPV/T) system for residential applications. | It used MATLAB/Simulink for dynamic modeling, combined with a neural network for irradiance prediction and a Non-dominated Sorting Genetic Algorithm II (NSGA-II) for multi-objective optimization. |
| 22 | Shahsavar et al. [90] | 2020 | It developed and compared multiple machine learning models to predict the exergetic performance of a building-integrated photovoltaic–thermal (BIPVT) collector based on design and operational parameters. | It used several machine learning techniques, including multiple linear regression, multilayer perceptron, radial basis function regressor, support vector machine (SMO), and lazy. IBK, random forest, and random tree to forecast the performance evaluation criterion (PEC). |
| 23 | Shahsavar et al. [100] | 2021 | The study developed predictive models to estimate the total annual energy output of a hybrid system combining a building-integrated photovoltaic/thermal unit and an earth-air heat exchanger. | It applied artificial neural networks (ANN), support vector machine networks (SVMN), and fuzzy networks (FN) to model and predict the system's performance. |
| 24 | Serrano-Luján et al. [47] | 2022 | The study developed a thermal prediction model for polycrystalline silicon BIPV modules using a hybrid artificial intelligence approach based on environmental and indoor parameters. | It employed a combination of Grammatical Evolution and Differential Evolution algorithms guided by the Sandia model structure. |

**Table A3.** *Cont.*

| No. | Reference | Year | What Did the Work Do? | What Method Did It Use? |
|---|---|---|---|---|
| 25 | Saw et al. [84] | 2022 | The study predicted the short-circuit current and simulated I-V parameters of colored BIPV modules using a hybrid approach combining machine learning and equivalent circuit modeling. | It used optimized machine learning models (SVR, GPR) to estimate short-circuit current, followed by one-diode equivalent circuit simulations to compute I-V characteristics. |
| 26 | Sankara Kumar et al. [89] | 2024 | The study forecasted the output power of a building-integrated photovoltaic (BIPV) system under different climatic conditions using environmental inputs. | It used a hybrid deep learning approach combining long short-term memory (LSTM) networks with Dragonfly and Firefly algorithms for feature selection. |
| 27 | Ragupathi and Ramasubbu [101] | 2022 | The study proposed a multi-time-scale forecasting model for predicting the power output of building-integrated photovoltaic systems. | It used a hybrid deep learning model based on long short-term memory (LSTM) networks optimized by a combination of Chicken Swarm Optimization (CSO) and Gray Wolf Optimizer (GWO). |
| 28 | Polo et al. [102] | 2023 | The study investigated hourly power forecasting for building-integrated photovoltaic (BIPV) systems on vertical façades using two years of monitored data from south- and east-facing arrays. | It applied gradient boosting (XGBoost) and random forest algorithms, including deterministic and probabilistic forecasting using the Python-based skforecast library. |
| 29 | Oviedo-Cepeda et al. [46] | 2021 | The study assessed the energy flexibility of a solar net-zero energy institutional building in Canada by optimizing the interaction between building technologies like BIPV, heat pumps, and thermal storage. | It used a machine learning algorithm to model thermal dynamics and a model predictive control approach with varying horizons (fixed, rolling, receding) to optimize energy management. |
| 30 | Nur-E-Alam et al. [93] | 2024 | This study designed and simulated an all-photovoltaic hybrid system integrating rooftop panels, solar windows, and semi-transparent PV modules to meet urban building energy demands while promoting sustainability. | The authors used HOMER Pro software for simulation and optimization of the hybrid system across multiple cities, supported by machine learning and IoT integration for enhanced energy efficiency. |
| 31 | Nguyen and Ishikawa [91] | 2023 | The study developed an artificial neural network model to predict the annual output energy of 4-terminal perovskite/silicon tandem photovoltaic cells for building-integrated photovoltaic applications. | The method combined simulated data from Atlas software with an optimized artificial neural network architecture, enhanced using a surrogate optimization algorithm. |
| 32 | Nguyen and Ishikawa [36] | 2022 | It predicted the annual output energy of 2-terminal perovskite/silicon tandem solar cells for building-integrated photovoltaics under realistic environmental conditions. | It used an artificial neural network (ANN) trained on Atlas-based simulation data. |
| 33 | Nguyen et al. [25] | 2024 | The study predicted the annual energy yield of perovskite/silicon tandem photovoltaic configurations (2T, 3T, 4T) under outdoor conditions for building-integrated applications in Japan. | It used five supervised machine learning models—regression trees, Gaussian process regression, ensembles of trees, support vector machines, and artificial neural networks—trained on simulated and measured data to perform the predictions. |
| 34 | Naeem and Fouad [92] | 2024 | The paper assessed the impact of environmental factors on photovoltaic performance in Cairo and proposed an intelligent system to enhance smart building sustainability by forecasting PV productivity. | It employed an artificial neural network (ANN) approach supported by empirical measurements and SCAPS/MATLAB simulations to forecast photovoltaic output under varying environmental conditions. |
| 35 | Maraveas et al. [103] | 2021 | The authors reviewed recent advancements and future perspectives on smart and solar greenhouse covers, emphasizing intelligent photovoltaic systems, material optimization, and IoT-based technologies for energy-efficient agriculture. | The authors reviewed recent advancements and future perspectives on smart and solar greenhouse covers, emphasizing intelligent photovoltaic systems, material optimization, and IoT-based technologies for energy-efficient agriculture. |
| 36 | Luo et al. [52] | 2020 | The study proposed and compared three machine learning-based multi-objective prediction frameworks for forecasting multiple building energy loads, including heating, cooling, lighting, and BIPV power generation. | It used artificial neural networks, support vector regression, and long-short-term memory neural networks for simultaneous multi-load prediction. |

**Table A3.** *Cont.*

| No. | Reference | Year | What Did the Work Do? | What Method Did It Use? |
|---|---|---|---|---|
| 37 | Liu et al. [85] | 2023 | The study proposed an optimal design method for photovoltaic shading devices (PVSDs) that combines geometric optimization with a machine learning–based adaptive control model to maximize energy and daylighting performance in office buildings. | It used EnergyPlus simulations for energy modeling, a treed Gaussian process (TGP) for sensitivity analysis, and a decision tree classifier (DTC) to develop the adaptive control model (ACM). |
| 38 | Zheng Li et al. [19] | 2025 | The study proposed a shadow-attention graph neural network to improve solar irradiation prediction for urban buildings in New York City and evaluated its implications for achieving net-zero energy buildings. | It used a novel Shadow-Attention graph neural network (SAGNN) to analyze interactions between buildings and predict solar irradiation with high spatial resolution. |
| 39 | Li and Ma [18] | 2024 | The study assessed large-scale photovoltaic potential on building roofs and facades in Manhattan by incorporating both into a city-wide solar irradiance model. | It used a geo-aware graph attention network (GA-GAT) to predict solar irradiation with high spatial resolution while accounting for building interactions. |
| 40 | Lee et al. [35] | 2020 | It proposed a method to improve short-term hourly power output predictions for building-integrated photovoltaics using machine learning and feature engineering. | It used a recurrent neural network (RNN) with feature engineering, including dropout observation and variable importance analysis via support vector machine (SVM). |
| 41 | Kabilan et al. [54] | 2021 | The study developed and validated short-term power prediction models for building-integrated photovoltaic (BIPV) systems across different building orientations using environmental data. | It used machine learning algorithms, specifically artificial neural networks (ANN), quadratic support vector machines (QSVM), and decision trees (TREE), to model and compare the predictive accuracy of PV power output. |
| 42 | Jouane et al. [30] | 2023 | This study developed and tested CNN, LSTM, and hybrid CNN-LSTM models to predict short-term photovoltaic energy production for a winter house using multivariate time-series data. | The method involved deep learning models—specifically CNN, LSTM, and a hybrid CNN-LSTM—applied to multivariate time-series forecasting with data from a Positive Energy Winter House. |
| 43 | Jeong et al. [53] | 2023 | The study proposed a machine learning approach to predict day-ahead hourly power output of building-integrated photovoltaics using weather forecast data and a novel feature called modified sky condition. | It used artificial neural networks combined with feature engineering, applying a new derived parameter, and evaluating prediction accuracy under various training and input scenarios. |
| 44 | Javadijam et al. [29] | 2024 | The study developed and optimized a thermoelectric-enhanced building-integrated photovoltaic thermal (BIPV/T) system for improved energy, exergy, and economic performance. | The work employed a hybrid method combining artificial neural networks (ANN), NSGA-II (Non-dominated Sorting Genetic Algorithm II), and TOPSIS for multi-objective optimization. |
| 45 | Samarasinghalage Tharushi Imalka et al. [44] | 2024 | The study developed a data-driven optimization framework to improve the design of building-integrated photovoltaic (BIPV) envelopes during the detailed design phase. | It used artificial neural networks (ANN) as surrogate models combined with the NSGA-II multi-objective optimization algorithm. |
| 46 | Hu and You [28] | 2024 | The paper evaluated the integration of AI-based robust model predictive control in large-scale photovoltaic-powered controlled environment agriculture within urban areas. | The study used year-round simulations in ten major U.S. cities, incorporating a cyber-physical-biological system framework that applied physics-informed deep learning and data-driven robust model predictive control. |

**Table A3.** *Cont.*

| No. | Reference | Year | What Did the Work Do? | What Method Did It Use? |
|---|---|---|---|---|
| 47 | Geng et al. [33] | 2025 | The authors developed a deep learning model using a multi-dimensional single-channel convolutional neural network to accurately predict building-integrated photovoltaic (BIPV) potential in dense urban areas based on LiDAR-derived 3D point cloud data. | They used a CNN model enhanced with surface sampling, normal estimation, Gaussian Mixture Model (GMM)-based WWR extraction, and feature fusion from building point cloud data and urban morphological parameters. |
| 48 | Gao et al. [41] | 2025 | The authors developed an interpretable deep reinforcement learning framework that integrates future forecast data and time-series network architectures to optimize the control of building-integrated photovoltaic and battery (BIPVB) systems. | The study used Soft Actor–Critic (SAC) reinforcement learning combined with GRU and Transformer neural networks for time-series processing, and SHAP (Shapley Additive Explanations) for interpretability. |
| 49 | Fara et al. [43] | 2021 | The study applied ARIMA and artificial neural network (ANN) models to forecast energy production of photovoltaic systems for both a lab-scale BIPV system and a large PV park in Romania. | It used statistical ARIMA models and machine learning-based ANN models, with accuracy comparisons and further enhancement using a solar radiation variability index. |
| 50 | Du et al. [32] | 2025 | The study developed a parametric modeling and optimization framework to analyze how urban block forms in Wuhan affect building energy use, urban heat island intensity, and photovoltaic potential. | The work used a multi-objective optimization approach integrated with machine learning (SVM), deep learning (CNN), and environmental simulation tools (Ladybug, Honeybee, and Dragonfly) within Rhino/Grasshopper. |
| 51 | Oukhouya et al. [104] | 2023 | The study proposed a new big data architecture for education systems that integrates data lakes to manage and consolidate heterogeneous data sources. | The method involved designing a layered architecture using data lakes and data warehouses to modernize data management and analytical processes in education systems. |
| 52 | Dimd et al. [38] | 2023 | The study evaluated how mixed orientations in building-integrated photovoltaic systems affect the accuracy of output power forecast models. | The authors used long short-term memory (LSTM) neural networks to train and compare forecasting models at different system levels. |
| 53 | Choi et al. [42] | 2024 | The authors developed a novel forecasting framework that combines Conditional GAN and TimeGAN to generate synthetic building-integrated photovoltaic (BIPV) power data, improving prediction accuracy and addressing data scarcity issues. | The study employed a hybrid Conditional GAN-TimeGAN approach with a tailored learning scheme that incorporates temporal conditions and specialized loss functions to generate high-fidelity time-series data for BIPV forecasting. |
| 54 | Chen et al. [105] | 2025 | The study developed a data-driven framework combining simulation, prediction, and optimization to design and assess academic building-integrated photovoltaic systems under climate change scenarios. | It used Light Gradient Boosting Machine (LGBM) for performance prediction and Adaptive Geometry Estimation MOEA (AGE-MOEA) for multi-objective optimization. |
| 55 | Chen et al. [105] | 2025 | The study developed a machine learning model to reconstruct visible and near-infrared components of solar radiation from broadband irradiance data for use in building-integrated solar energy simulations. | It used the Extreme Gradient Boosting (XGBoost) regression method along with SHAP for feature analysis. |

**Table A3.** *Cont.*

| No. | Reference | Year | What Did the Work Do? | What Method Did It Use? |
|---|---|---|---|---|
| 56 | Biloria et al. [45] | 2023 | The study investigated the performance of a real-time adaptive BIPV (building-integrated photovoltaic) shading system in enhancing both energy generation and visual comfort in a multistorey building in Sydney. | It employed a multi-objective evolutionary algorithm (MOEA), specifically NSGA-II, integrated with simulation tools in Rhino/Grasshopper, to optimize BIPV panel orientation based on solar irradiance and interior illuminance metrics. |
| 57 | Baz and Patel [40] | 2024 | The study designed and analyzed two solar absorber structures (single-layer and multilayer) using graphene and metamaterials, and optimized their performance for solar thermal applications. | The works used COMSOL Multiphysics simulations combined with machine learning (random forest regression) for parametric optimization and absorption prediction. |
| 58 | Asghar et al. [106] | 2024 | The study evaluated and compared the performance of ANN, LSTM, GRU, and CNN models for forecasting BIPV power generation in Rome using a six-year dataset. | The authors employed standardized deep learning algorithms—ANN, LSTM, GRU, and CNN—trained and tested using PVGIS weather data, with performance assessed via RMSE, MAE, and $R^2$ metrics. |
| 59 | Alsagri and Alrobaian [27] | 2024 | The study analyzed and predicted the performance of a building-integrated photovoltaic/thermal (BIPV/T) system with and without a phase change material layer, using simulations and machine learning for the years 2022, 2024, and 2025. | It employed TRNSYS and MATLAB for simulations, and random forest machine learning in Python for performance prediction and weather data forecasting. |
| 60 | Bosu et al. [56] | 2023 | The study conducted a comprehensive review of single and hybrid solar energy techniques—both passive and active—such as solar chimneys, Trombe walls, and photovoltaics, highlighting their roles in improving energy efficiency and thermal comfort in buildings. | The authors used a literature review approach, analyzing theoretical, numerical, and experimental studies on solar energy applications in buildings. |
| 61 | Abouelaziz and Jouane [107] | 2024 | The study developed a novel method called BIM-AITIZATION that integrates photogrammetry, BIM data, and deep learning to predict photovoltaic energy production and support decarbonization in building-integrated photovoltaics (BIPV). | It used a combination of photogrammetric point cloud processing and a convolutional neural network (CNN) model trained on meteorological and BIM parameters to automate and enhance the accuracy of BIPV energy prediction. |

Abbreviations: ANN—artificial neural network; CNN—convolutional neural network; LSTM—long short-term memory; RF—random forest; KPI—key performance indicator; MAE—mean absolute error; RMSE—Root Mean Square Error.

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